BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *Autonomous Library Book Retrieval Robot (LibBot)*
by

Group_No: 7
Section: 8

Group Members:

Name	ID
Wasi Farabi	22201358
Swapnil Rob	22299138
Md Iftekher Ahmed	22301631
Sayeda Ariyan Fairooz	23201480

Department of Computer Science and Engineering
BRAC University
4 September, 2026

Abstract

LibBot is a prototype scale, autonomous robot designed for book retrieval in a library to demonstrate how robotics, mobile line-following robot navigation, obstacle avoidance capability, RFID identification capability and robotic manipulation can be integrated into one embedded system. The central controller will be an Arduino Uno microcontroller, the L298N dual H-bridge driver will control two geared DC motors for differential drive, a five channel infrared line sensor array will be used for path tracking, an HC-SR04 ultrasonic sensor for obstacle detection, an RC522 RFID reader for shelf and collection point identification and finally a four-servo robotic arm for book pick up and release. A regulated power architecture isolates the motor supply and 5V logic/servo rail, allowing a common electrical ground. The final prototype is able to complete the tasks of line following, obstacle-aware movement, RFID based shelf confirmation, robotic arm manipulation, book retrieval, return navigation, and delivery to a designated collection point. The project demonstrates how a low cost embedded platform can integrate sensing, decision making, locomotion, and manipulation to solve repetitive library retrieval tasks in a controlled indoor environment.

Keywords

Autonomous Robot, Library Automation, RFID, Line Following, Arduino Uno, Robotic Arm, HC-SR04, L298N

1. Introduction

Retrieval of books from shelves is repetitive and time consuming, especially in a library that has a large number of books on shelves or users who cannot reach or search for books on their own. This proposal was thus to create LibBot, an independent robot that can pick up a book when it is requested, move through a mini library layout, locate the shelf where the book is stored, retrieve the book with the robotic arm, and place the book in a collection point. The prototype is designed to be used in a controlled environment, indoors, where the paths are pre-planned, the shelves are marked, and the demonstration books are lightweight.

This implementation is a simplified version of the original proposal to a very solid single-controller version. The completed robot has a single line sensor, a single Arduino Uno, the motors are replaced with two DC motors, the multiple ultrasonic sensors are replaced with a single HC-SR04 sensor, the multiple motor drivers are replaced with one L298N motor driver, and the three active channels of the five-channel line sensor array are used. The lowered amount of hardware equipment decreases integration complexity, but still maintains the three functional components: autonomous navigation, precise identification of the shelves, and retrieval and delivery of books.

2. Related Work/ Inspiration

LibBot is inspired by three existing fields of robotics: automated guided vehicles (AGVs), identification systems using radio frequency identification (RFID), and robotic manipulators. A typical AUV system in a factory or warehouse environment is one that is marked. LibBot takes this concept a step further and follows a black line and counts the junctions in a miniature library. RFID is used extensively for contactless identification and inventory. For this project, location confirmation that is as precise as possible after coarse navigation by lines, is achieved by using unique shelf tags. The final LibBot includes four independently controlled servo joints for positioning the arm and the gripper and picking up a small book, then pulling it back into the bot and dropping it at the collecting location.

The amazing factor about LibBot is that all these techniques are brought together into a neat, inexpensive educational system. The robot uses a state based sequence of control and deterministic sensors to avoid using computer vision which can be computationally expensive. This simplifies the system to be more easily understood, tested, troubleshooted, and shown within the scope of this course.

3. Technical Approach

The project was designed by integrating the various subsystems (sensing, navigation, identification, motor-control, robotic arm and power) into a unified platform. The Arduino Uno is the primary controller which receives data from the sensors and sends commands to the wheels, RFID reader, and servo motors.

3.1 System Architecture:

The system is split into sensor, decision and control, locomotion, identification, manipulation and power sub-systems. The data from IR line sensor array, HC-SR04 sonar, and RC522 RFID reader are given to the Arduino Uno. It then sends the commands to the motor driver L298N and the four servo motors depending upon the operating condition. The sensing subsystem enables the robot to determine its local surroundings and its location to the specified track. The line following is done by IR sensors, the obstacle detection is done by an ultrasonic sensor which is placed in front of the robot and the exact position of the robot on the shelf or collection point is checked by the RFID reader. The locomotion subsystem is made up of the two DC geared motors driven via the L298N. The four-servo robotic arm and gripper make up the manipulation subsystem.

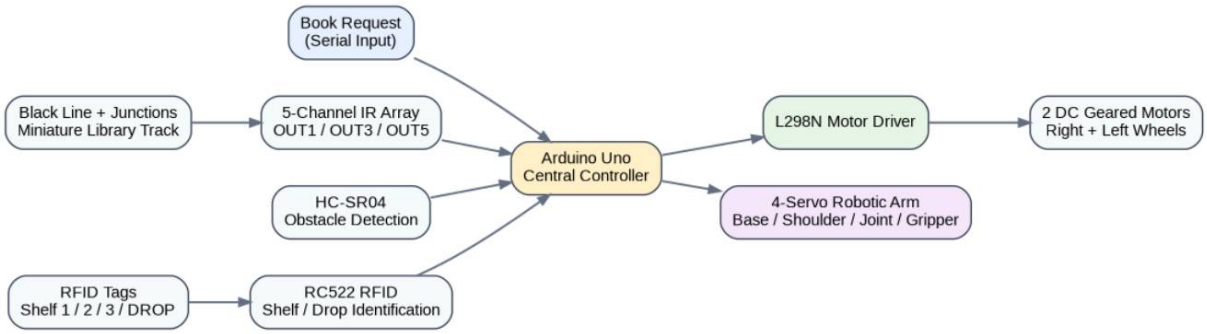


Fig. Block Diagram

System Workflow:

A user or librarian requests a book using the serial interface first. LibBot then starts to follow the marked path and constantly calculates the ultrasonic distance and counts the line junctions to go to the right aisle. After this point, the RC522 reader reads the RFID tag in its vicinity. If the RFID tag detected is not the requested shelf, then the robot will not do the pick-up operation. Rather, it keeps on scanning until it locates the appropriate tag. When the correct tag is detected, the four-servo robotic arm starts the book-pickup sequence. Once the book is in place the robot follows the line back toward the book collection point.Once the book is in place, the robot will follow the line back to the book collection point. As soon as the dedicated tag is detected, the arm adopts the release position and then drops the book at the collection area. Once the robot finishes the task, it goes back to its idle state and is ready for the next request.

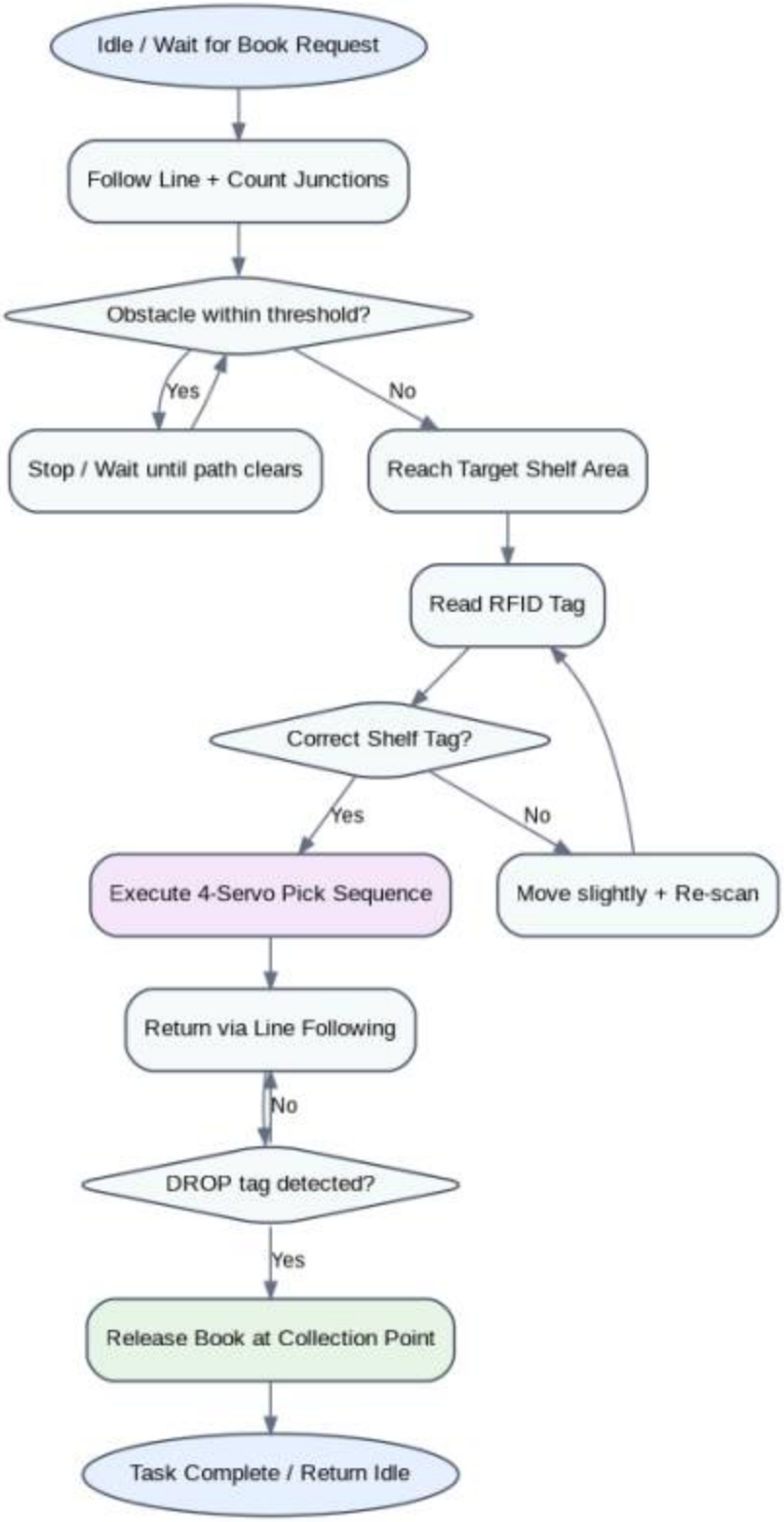


Fig. Workflow Diagram

3.2 Components Used:

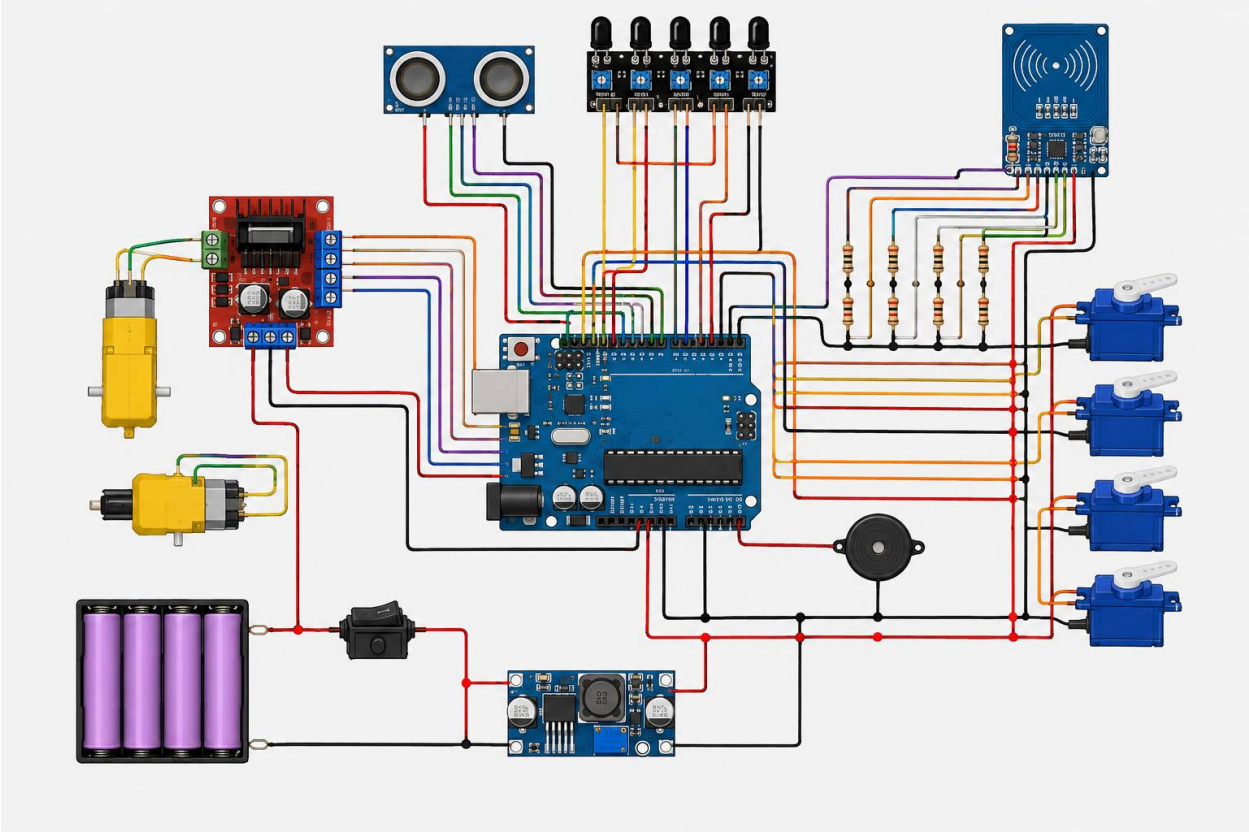
Component	Description
Arduino Uno	Central controller for sensors, RFID, motors and four servos
L298N motor driver	Dual H-bridge control of right and left DC wheel motors
DC geared motors + wheels	Differential-drive locomotion
5-channel IR line sensor array	Line Tracking
HC-SR04 ultrasonic sensor	Forward obstacle detection
RC522 RFID reader	Shelf and collection-point identification
13.56 MHz RFID tags/cards	Unique identifiers for three shelves and DROP point
Four-servo robotic arm	Base, upper joint(s) and gripper manipulation
LM2596 buck converter	Regulated 5.0 V supply for logic/sensors/servos
Rechargeable battery pack	Main energy source
Acrylic 2WD chassis + caster	Mechanical platform
Breadboard, jumper wires, resistors	Prototyping, power distribution and RFID voltage dividers

3.3 Cost Breakdown:

Component	Unit Cost	Quantity	Total
Arduino Uno	1000	1	1000
L298N H-Bridge dual motor driver	220	1	220
HC-SR04 ultrasonic sensor	100	1	100

Line Following Sensor 5 Array	350	1	350
RC522 RFID reader kit	200	1	200
Additional MIFARE 13.56 MHz cards/tags	32	5	160
3D Printed Arm+Gripper	700	1	700
2WD acrylic chassis kit with 2 motors/wheels/caster	645	1	645
LM2596 buck converter module	250	1	250
Battery Pack	500	1	500
Servo Motors	180	4	720
Misc (Wires, Breadboard etc)	450	1	450
Total	-	-	5295

3.4 Circuit/Schematic:



3.5 Functionality:

Autonomous Navigation and Obstacle Avoidance:

LibBot moves around in the five channel IR array with left, centre and right channels. The centre sensor is placed close to the middle of the robot and the outer sensors are used to find out if the robot is moving away from the path. The Arduino reads the sensor values and transmits the appropriate instructions to the L298N motor driver. The motor driver then uses the two DC motors to ensure that the robot stays on the track. The HC-SR04 ultrasonic sensor is used to continuously monitor the space in front of the robot and a buzzer is to alert obstacle detection. When an object is within the chosen safe distance, the robot stops rather than advances towards the object. When the path is clear, normal navigation resumes. The line sensor system also detects junction bars on the line. These junctions give the coarse localization information and let the robot know when it is in the area of the requested shelf.

RFID Shelf and Collection-Point Identification:

RFID serves as the second navigation and identification layer. Shelves with the books and the book drop or collection point all have unique RFID tags. The RC522 reader is attached to the side of the robot that is facing toward the shelf tags. The LibBot reads the nearby RFID tag when it enters the target shelf region.

If the tag is not on the requested shelf, the robot doesn't try to get the book. Rather, it does not stop until it finds the right tag. As soon as the matching tag is found, the robot will stop at that tag and will begin the arm-control sequence. This way will reduce the chance that the robot will pick up a book from an incorrect shelf.

Four-Servo Book Retrieval and Delivery:

The last retrieval mechanism is a four degree-of-freedom servo arm attached to the mobile chassis. Servo 1 is for the base of the arm. Servo 2 controls the joint that is directly above the base. Another arm joint is controlled by Servo 3, and the gripper is controlled by Servo 4. Once the proper RFID shelf tag has been verified, the Arduino begins to send a series of servo-angle commands. The arm moves the gripper forward in front of the book, it then closes the gripper around the book and pulls the book off the shelf. When the book is retrieved, the robot heads back to the collection point but maintains the arm in a safe carrying position. The final manipulator is modified from the previous project proposal in that instead of a simple 2 servos, the finished bot now has 4 servos motors.

3.6 Challenges:

We encountered several challenges during this project of LibBot. Controlling the 3D printed arm and the gripper remains very difficult. As it needs very precise servo movement and alignment which sometimes become unpredictable and work randomly. Moreover, line following was sometimes challenging as the 5-array IR sensor sometimes detected shadows as lines and deviated from the path. Furthermore, controlling the DC geared motors were challenging due to the variations in motor speed and response. It made it difficult to achieve consistent turns and movements. Thus, we faced a lot of challenges while implementing this project.

4. Sustainability & Impact

4.1 Sustainability:

This is seen as an approach to small-scale robotics that is reparable and reusable in LibBot. Arduino, Sensors, Motor Driver, DC Motors, Servo Motors and the Chassis can be individually replaced when they fail. Rechargeable batteries eliminate reliance on single-use batteries. The robot also utilizes a relatively simple technique of sensor based control instead of using complicated hardware to assist in keeping power requirement and overall component cost relatively low. The mechanical structure can also be adapted and/or remounted for future robotics projects so that the prototype can be used for further educational development rather than for one-off construction.

4.2 Impact:

Automated Library Book Retrieval System can decrease repetitive walking and searching tasks in the huge libraries for the librarians. A robot would be useful in a larger implementation to help the user by

delivering books they have requested to a convenient collection point. This could also benefit those who are physically unable to move about extensively in a large library or who have trouble accessing certain books and shelves, including older people. The prototype shows that low-cost embedded hardware can be integrated to realize an assistive service robot capable of sensing, autonomous locomotion, identification, and physical manipulation.

4.3 Future Work:

- Install an additional controller or I/O expander to free up extra pins for extra confirmation sensors and user interfaces.
- Connect a library database to automatically map a book ID to the shelf coordinates and availability.
- Use wireless request input via ESP32, Wi-Fi / Bluetooth and a simple user-friendly app.
- Introduce charging-dock localization and automatic return-to-charge behavior.
- Enhance the gripper geometry, with force and limit sensing for closed-loop pick up verification.

4.4 Limitations:

- The practical reading distance of the RC522 is short, which means that the tag needs to be positioned continuously.
- The four-servo 3D printed arm is designed for lightweight demonstration books, heavier books are beyond the ability of the gripper/servo.
- There is not much memory and I/O space left on the Arduino Uno, as there are already all the pins used in the final pin map, with no additional sensors to be added without redesigning the board.
- The current system is based on deterministic shelf/tag mapping; dynamic library inventory integration is not included in the prototype.

5. Results & Discussion

Navigation: Follows the black line using 5-array IR sensor

Obstacle Avoidance: When an obstacle is detected the robot stops and buzzer is turned on

Shelf Identification: RFID works to detect the target book shelf

Book Retrieval: Arm and gripper controlled by servo motors pick the book and release

Overall Result: The prototype demonstrated the integration of sensing, navigation, actuation and control for the book retrieval robot

6. Conclusion

The LibBot is an Autonomous Library Book Retrieval Robot that successfully demonstrates a complete task using low cost embedded hardware. The overall design is a single Arduino Uno controlling line following, obstacle detection, RFID shelf verification, differential drive movement and a 4 servos robotic

arm. The final design ended up being quite different from the initial design that was more of a broad concept. The number of controllers was decreased and some hardware components were removed, but the main project functions were kept. This simplified the architecture and also made the entire system simpler to incorporate and also troubleshoot. Once the robot is finished, it can pick up a book request, move toward the book, verify the exact location by RFID, use the robotic arm to pick up the book, go back to the collection point, and then drop the book. The project thus shows the successful integration of sensing, embedded control, autonomous locomotion, identification, and robotic manipulation. It also demonstrates the potential of low-cost robotics for library-assisted applications with controlled robot behavior.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Wasi Farabi	22201358	Hardware & Control Integration	● Designed and assembled the overall robot hardware architecture ● Worked on the 3D-printed robotic arm and gripper mechanism ● Programmed motor and servo control functions ● Assisted with integrating the navigation, RFID, and book-retrieval mechanisms
Swapnil Rob	22299138	Navigation & Sensor Integration	● Implemented and calibrated the 5-array IR sensor for line following ● Worked on ultrasonic sensor for obstacle detection and avoidance ● Assisted in developing navigation and movement logic ● Tested and optimized sensor-based movement
Md Iftekher Ahmed	22301631	Mechanical Integration	● Integrated the RFID reader for shelf/book identification ● Integrated DC motors, motor driver, servos, battery, and buck converter ● Managed overall component placement and wiring ● Worked on Arduino control logic for sensor and actuator coordination

Sayeda Ariyan Fairooz	23201480	Software, Testing & Documentation	• Developed and organized the Arduino control code • Worked on logic for sensing, movement, book retrieval, and delivery • Tested individual components and debugged system-level issues • Prepared project documentation, report materials

References

MArobotic. (2018, May 5). *RFID and line follower based automatic book picking robot for libraries | Library robot* [Video]. YouTube. https://www.youtube.com/watch?v=tHyCp647rsc

Yu, X., Fan, Z., Wan, H., He, Y., Du, J., Li, N., Yuan, Z., & Xiao, G. (2019). *Positioning, navigation, and book accessing/returning in an autonomous library robot using integrated binocular vision and QR code identification systems.* Sensors, 19(4), 783. https://doi.org/10.3390/s19040783